# Unit XVI: Proteins (with Optical Isomerism)

Lecture notes: the slides plus what she explained in class (transcript)

CHEMISTRY (PROF. L. RANAY)

## WHY BIOMOLECULES MATTER (HER OPENING)

- Biomolecules react and transform to give us all the **characteristics of a living thing** (movement, adaptation/homeostasis, reproduction), not only energy.
- As future doctors, the one goal of everything you study is **continuity of life**; metabolism exists to keep these characteristics going.
- Four major biomolecules: **proteins, nucleic acids, lipids, carbohydrates**. **Only proteins, nucleic acids and carbohydrates are macromolecules** (they have building blocks); **lipids are not**.

## PROTEINS

- **Naturally occurring, unbranched polymer** whose monomers are **amino acids**. Unbranched = a straight chain with no branches at all (DNA is also unbranched).
- **Most abundant substance in cells after water**: about **15%** of a cell's overall mass.
- Elements: **C, H, N, O and S**. Average **nitrogen** content: **15.4% by mass**.
- **The body cannot store nitrogen** (it can store carbohydrates and lipids/fat), so excess nitrogen is excreted as nitrogen wastes: **urea, uric acid, ammonia** (urine).
- **Fe, P and other metals** occur only in certain **specialized** proteins; they are additional, not part of the basic protein structure.

## AMINO ACIDS

- Contain both an **amino group ($-NH_2$)** and a **carboxyl group (–COOH)**: that is why they are called amino acids.
- **α-amino acids:** the amino and carboxyl groups are attached to the **α-carbon**. General structure: $H_2N–CH(R)–COOH$.
- **Side chains (R)** differ in size, shape, charge, acidity, functional groups, H-bonding ability and reactivity. **More than 700** amino acids are known.
- **Standard amino acids:** the **20** α-amino acids normally found in proteins; they must be present in the body. Others (e.g., **hydroxyproline**) can be absent.
- **Essential amino acids** must come from the **diet** (e.g., eggs, meat); **nonessential** ones the body can make.
- Three-letter and one-letter codes are shortcuts. Biology link: **three nucleotides (a codon) code for one amino acid**.

## FOUR CLASSES BY SIDE-CHAIN POLARITY

- **Look only at the side chain (R).** Every amino acid has the amino and acid groups, so ignore them when classifying.
- **1. Nonpolar (hydrophobic):** side chains are mostly **hydrocarbons**; **not attracted to water**; hide in the **interior** of the folded protein. **Nine**: **glycine, alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan**.
  - **Methionine** has S, but sulfur has low electronegativity (S and C are both **2.5**), so a C–S–C side chain is nonpolar.
  - **Tryptophan** has an N, but its large ring (nonpolar part) **overpowers** the single nitrogen.
  - "Hydrophobic" describes the behavior **toward water**; water strongly shapes protein structure.
- **2. Polar neutral:** polar but **neither acidic nor basic** at physiological pH; **more soluble in water** (R group can H-bond with water). **Six**: **serine, cysteine, threonine, tyrosine, asparagine, glutamine**.
  - Ser, Thr, Tyr: **–OH** (alcohols are polar but neutral). **Cys: –SH**; S next to H (2.5 vs 2.1) gives slight polarity. **Asn, Gln:** carbonyl O plus N (amide).
- **3. Polar acidic:** a **–COOH in the side chain**. **Two**: **aspartic acid, glutamic acid** (names end in "acid").
- **4. Polar basic:** an **amino group (N) in the side chain** (amines are bases). **Three**: **lysine, histidine, arginine**.

**Note:** Classification is of the **side chain only**. As a whole molecule, **glycine is polar** (it has the amino and carboxyl groups), but its side chain (H) puts it in the **nonpolar** class.

**Slide exercise**

(a) R = $CH_3$ → **nonpolar**; (b) R = $CH_2OH$ → **polar neutral** ("all with OH"); (c) R = $CH_2COOH$ → **polar acidic**; (d) R = $CH_2$–ring → **nonpolar**; (e) R = $(CH_2)_4NH_2$ → **polar basic**.

## OPTICAL ISOMERISM (OPTICAL ISOMERISM SLIDES)

- **Optical isomers** = a type of **stereoisomer**: **same molecular formula, same connectivity**, but different **3-D arrangement**, so they are **non-superimposable mirror images**. You see it only in 3-D, not on paper.
- **Chiral** objects have left- and right-handed forms; their mirror image is **non-superimposable**. Your **hand** is chiral: however you turn your right hand, it never matches your left hand (cutting a thumb off would be breaking bonds, a chemical reaction, and isomers involve no reaction).
- **Achiral** objects have **superimposable** mirror images (e.g., a **chair**).

### CHIRAL CENTER

- A **chiral center** (chiral carbon) = a carbon bonded to **four different groups**. **No double bond** on it (it needs four single bonds). A **ring** carbon can be chiral if it has four different groups.
- If a molecule has **at least one** chiral carbon it is **chiral**; with **no** chiral carbon it is **achiral**.
- Her example, **2-bromobutane:** C1 ($CH_3$, three H) and C3 ($CH_2$, two H) and C4 cannot be chiral; **C2** has $CH_3$, $CH_2CH_3$ (ethyl), H and Br = four different groups → **chiral**.
- If only three groups differ (two are the same, e.g., two H), the carbon is **not** chiral.

### ROTATING LIGHT: DEXTRO AND LEVO

- One isomer rotates plane-polarized light **clockwise** = **dextrorotatory (+), "d"/dextro, to the right**; the other **counterclockwise** = **levorotatory (-), "l"/levo, to the left**.
- Measured with a **polarimeter** in the lab. **R and S** are a separate **paper (IUPAC) designation**; don't mix them up with d/l. (She will **not** ask you to determine R/S or levo/dextro.)
- Her examples: **levocetirizine** (levo); **dextrose** = dextro-glucose: the body accepts only **D-glucose**, so giving a patient levo-glucose would be fatal; **methamphetamine**: d-isomer = "meth", l-isomer = in **Vicks** inhaler; **lactic acid** (+) from **muscle** (strenuous exercise, anaerobic) vs (-) from **milk**; **limonene** from lemons vs oranges.
- Why it matters: enzymes are very specific (**lock-and-key** and **induced-fit** models); a molecule that does not fit the enzyme does not react.

### COUNTING AND COMPARING STEREOISOMERS

- One chiral center → **2** enantiomers. With **n** chiral centers → up to $\mathbf{2^n}$ stereoisomers (n = 3 → **8**).
- **Enantiomers:** a pair of optical isomers that are **non-superimposable mirror images**: same formula, same connectivity, **same physical properties** (mp, bp, density); they differ only in rotating plane-polarized light in **opposite directions**.
- **Diastereomers:** stereoisomers that are **not mirror images** and not superimposable. Possible only with **two or more chiral carbons**. They have **different physical and chemical properties** (bp, solubility, reactivity). Example: (2S,3S) vs (2S,3R): same at C2, opposite at C3.
- Enantiomers and diastereomers are the **two kinds** of optical isomers she named.

## CHIRALITY OF AMINO ACIDS

- All standard amino acids have **four different groups on the α-carbon** (H, $NH_2$, COOH, R) **except glycine** (its R is **H**, so two H's) → glycine is the **only achiral** standard amino acid.
- So every other amino acid exists as L and D forms (e.g., L- and D-lysine, L- and D-methionine).
- **Proteins and amino acids in nature are L isomers** ("L for life"); **sugars are D** (dextrose).
- **Fischer projection rules** for amino acids: **–COOH at the top, R at the bottom, $-NH_2$ on the horizontal**: **left = L, right = D**.

**Slide exercise**

Name with the correct designation: A = **L-isoleucine**, B = **D-cysteine**, C = **L-tyrosine** (look at which side the $NH_2$ is on).

## ACID-BASE PROPERTIES

- An amino acid has an **acidic group (–COOH)** and a **basic group (–$NH_2$)** on the same carbon → **amphoteric** (can act as acid or base).
- In neutral solution the carboxyl group **loses** $H^+$ (→ $COO^-$) and the amino group **gains** $H^+$ (→ $NH_3^+$). No amino acid exists in the body as neutral COOH/$NH_2$.
- **Zwitterion:** a molecule with a **+** charge on one atom and a **-** charge on another but **no net charge**.
- Three forms are in equilibrium, and the equilibrium **shifts with pH**.
- **Isoelectric point (pI):** the pH at which an amino acid exists mainly as the **zwitterion**. The 15 nonpolar and polar neutral amino acids have pI **4.8 to 6.3**; the **3 basic** ones have **higher** pI, the **2 acidic** ones **lower**.
- **pH < pI:** excess $H^+$ protonates the carboxylate → net **positive**. **pH > pI:** excess $OH^-$ removes $H^+$ from the $–NH_3^+$ → net **negative**.
- Different parts of the body have different pH (stomach acidic, small intestine basic, mouth neutral), so **enzymes have different working pH**.
- Used in **gel electrophoresis** (amino acids move by their charge); electrophoresis is also used in **DNA fingerprinting**.

## CYSTEINE: THE CHEMICALLY UNIQUE AMINO ACID

- Side chain has a **sulfhydryl (–SH, thiol)** group.
- Two cysteines **dimerize** (oxidation) into **cystine**: the two residues are linked by a covalent **disulfide bond (–S–S–)**.

## PEPTIDES

- Amino acids link into a **peptide** (unbranched chain). The bond is the **peptide bond** (covalent); its functional group is the **amide**.
- Formed between the **amino group** of one amino acid and the **carboxyl group** of another.
- **Dipeptide** = 2 amino acids, **1** peptide bond; **tetrapeptide** = **3** peptide bonds (n amino acids → n - 1 bonds). **Oligopeptide** = 10 to 20 residues; **polypeptide** = long unbranched chain.
- Every peptide has an **N-terminal end** and a **C-terminal end** (you'll meet them again in protein synthesis).

## FOUR LEVELS OF PROTEIN STRUCTURE

### PRIMARY

- The **sequence** (order) of amino acids, i.e., the order of attachment by peptide bonds. The **peptide bonds** are the backbone and the most important bond here. Same wherever the protein is found in the organism.
- Specifying it means giving **the number and sequence of amino acids** (slide question, answer c).
- The sequence dictates the higher structures; **one wrong amino acid can cause disease**: **sickle cell anemia** = **glutamate replaced by valine at position 6 of the β-globin chain**; red cells become rigid, sticky, sickle-shaped. Sequence diseases are **genetic** (passed on; why doctors ask about family history).
- The six atoms of the peptide linkage lie in **one plane**; the backbone C and N atoms form a **zigzag**.

### SECONDARY

- The arrangement in space of the **backbone**: **α-helix** and **β-pleated sheet**.
- Held by **hydrogen bonds between the C=O oxygen of one peptide linkage and the N–H hydrogen of another** (backbone only, not R groups). Slide question: answer **a**.
- **α-helix:** coiled spring, **right-handed** (clockwise) twist; **intramolecular** H bonds **parallel to the helix axis**; each C=O bonds to the N–H **four residues** along; **3.6 residues per turn**; R groups point **outward**. She: found in **structural proteins** such as keratin; nonpolar side chains tuck inside, polar ones face the water.
- **β-pleated sheet:** two extended segments side by side, held by H bonds, either in one chain that folds back (**intrachain**, needs **U-turns**) or between different chains (**interchain**); zigzag "pleats"; H bonds lie **in the plane** of the sheet; R groups **above and below** it.
- **Unstructured segments** have neither. α-helix and β-sheet occur only where the R groups are **small**; **large R groups disrupt** them. Not every protein has both.

### TERTIARY

- The overall **3-D shape** from **interactions between side chains (R groups)**. (Secondary = backbone

C=O with N–H; tertiary = side chain with side chain.)

- **1. Disulfide bonds:** two cysteine –SH → covalent S–S; the **strongest**; intra- or intermolecular.
- **2. Electrostatic attractions = salt bridges:** an **acidic** R group ($-COO^-$) with a **basic** R group ($-NH_3^+$).
- **3. Hydrogen bonds:** between **polar** R groups; weak, disrupted by pH changes.
- **4. Hydrophobic interactions:** between **nonpolar** side chains (alkyl, phenyl); **London dispersion forces**; weaker than H bonds or salt bridges but significant because there are so many.

**Slide exercise**

Glycine and asparagine: **hydrogen bonding**. Glutamic acid and lysine: **electrostatic attraction** (salt bridge).

- **Denaturation** (her examples): **alcohol** (–OH) disrupts the **hydrogen bonds** in bacterial proteins. **Hair** is mostly nonpolar amino acids held by **disulfide bonds**, so alcohol does not dissolve it.
- **All proteins have up to tertiary structure; not all have quaternary.**

## QUATERNARY

- Only in proteins with **more than one polypeptide chain**; each chain is a **subunit**. **Oligomers** = molecules made of several subunits (dimers, trimers, tetramers).
- Subunits are held **noncovalently**: electrostatic attractions, hydrogen bonds, hydrophobic interactions.
- **Allosteric:** in a multi-subunit protein, a shape change in one subunit causes a large change in another.
- Possible for **multimeric** but not monomeric proteins (slide question, answer b).

## HEMOGLOBIN VS MYOGLOBIN

- **Hemoglobin: tetramer**, **two α chains (141 residues)** + **two β chains (146 residues)** = $\alpha_2\beta_2$; each chain has a **heme** group with **iron** at the center → **4 $O_2$** per hemoglobin; carries $O_2$ from the lungs to tissues; has quaternary structure.
- **Myoglobin: one** chain + one heme → **one $O_2$**; **oxygen storage in muscle**; **no quaternary structure**; **higher $O_2$ affinity** than hemoglobin (so it does not release $O_2$ easily: unsuitable as the blood's carrier).
- Most amino acids of the α chain, β chain and myoglobin are **homologous**.

## CLASSIFYING PROTEINS

- **Protein** = a peptide with **at least 40** amino acid residues; "protein" and "polypeptide" are used interchangeably. Small proteins: 40 to 100 residues.
- **By number of chains: monomeric** (one chain) vs **multimeric** (more than one; the chains are **subunits**).
- **By composition: simple** (only amino acids) vs **conjugated** (also a non-amino-acid part = the **prosthetic group**).

## SUMMARY TABLE: CONJUGATED PROTEINS

| Class | Prosthetic group | Example (function) |
|---|---|---|
| Hemoproteins | heme | hemoglobin ($O_2$ carrier in blood), myoglobin ($O_2$ binder in muscle) |
| Lipoproteins | lipid | LDL, HDL (lipid carriers) |
| Glycoproteins | carbohydrate | gamma globulin (antibody), mucin (lubricant in mucus), interferon (antiviral) |
| Phosphoproteins | phosphate | glycogen phosphorylase (enzyme) |
| Nucleoproteins | nucleic acid | ribosomes (protein synthesis), viruses |
| Metalloproteins | metal ion | ferritin (iron storage), alcohol dehydrogenase (zinc; alcohol oxidation) |

- She corrected herself in class: myoglobin is **conjugated**, not simple (it has a heme with iron).
- **By shape: fibrous** = elongated, one dimension much longer, linear chains that aggregate;

**structural**. **Globular** = chains folded into a **spherical** shape, **hydrophilic side chains outside, hydrophobic inside**. **Enzymes are globular**: they must be flexible to fit every reaction (a fibrous enzyme could not work).

## WHAT TO REVIEW

- Protein facts (unbranched, after water, 15%, CHONS, 15.4% N, no N storage → urea).
- The four side-chain classes with members, and how to classify from a structure (side chain only; glycine trick).
- Chirality: chiral center rule, 2-bromobutane, glycine achiral, L amino acids vs D sugars, Fischer L/D, enantiomers vs diastereomers, $2^n$.
- Zwitterion, pI, charge above/below pI; cysteine → cystine (disulfide).
- Peptide bond = amide, n - 1 bonds, N- and C-terminal.
- The four levels with their bonds/forces; sickle cell; α-helix vs β-sheet details; the four tertiary forces; denaturation examples; quaternary, hemoglobin vs myoglobin.
- Classification by chains, composition (conjugated table) and shape.